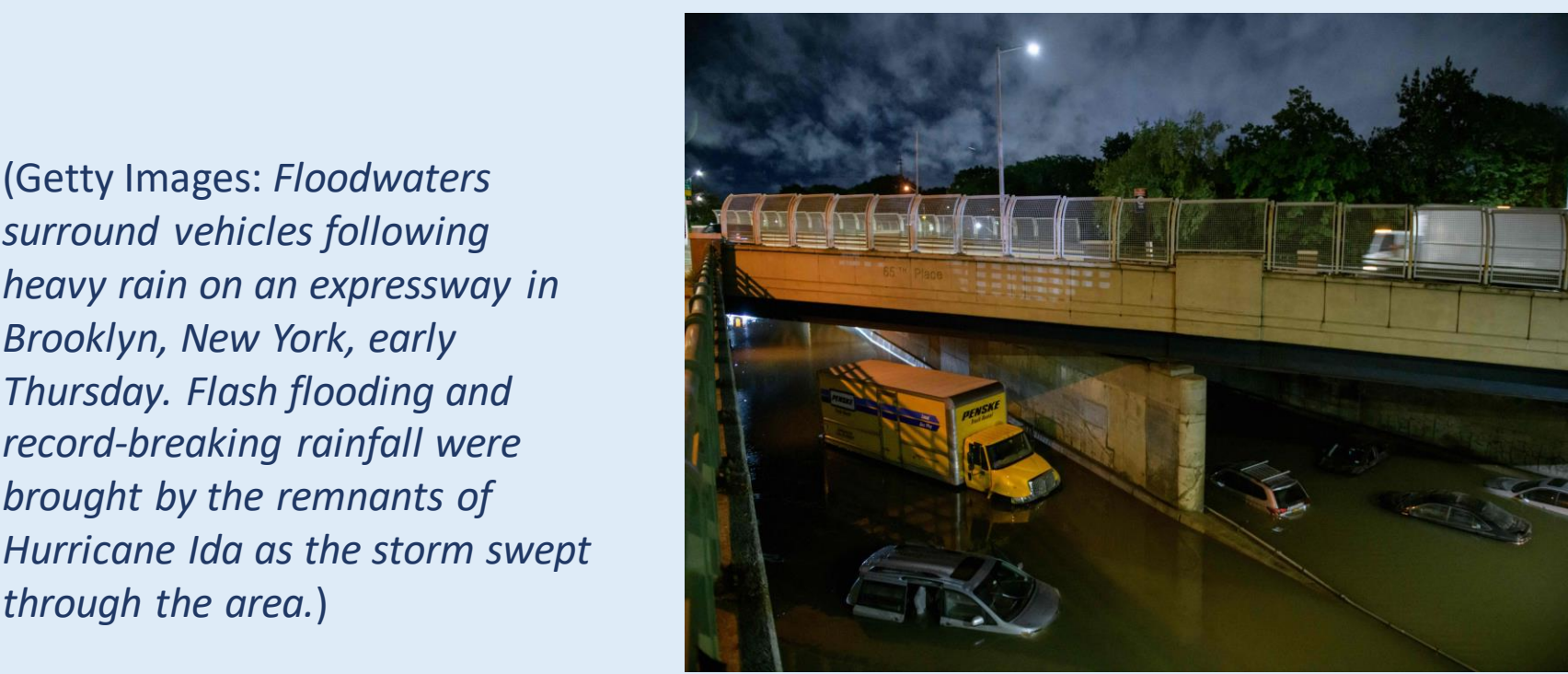


INTRODUCTION:

- The remnants of Hurricane Ida were devastating to the New York area.
- 11 fatalities in NYC were caused by rapid flooding in basement homes (Yam and Venkatraman, 2021).
- Understanding flood distribution is key to understanding flood vulnerability.
- The 311 system is a government run resource service that provides assistance and information to New Yorkers outside of emergency situations.
- All calls made to 311 are recorded in an open-source database with their request type, time, and location.
- 311 calls related to flooding can be used as a way to measure relative flood vulnerability.



(Watts, Anna: New Yorkers tried to save neighbors from Ida's floods. Sometimes they couldn't.)



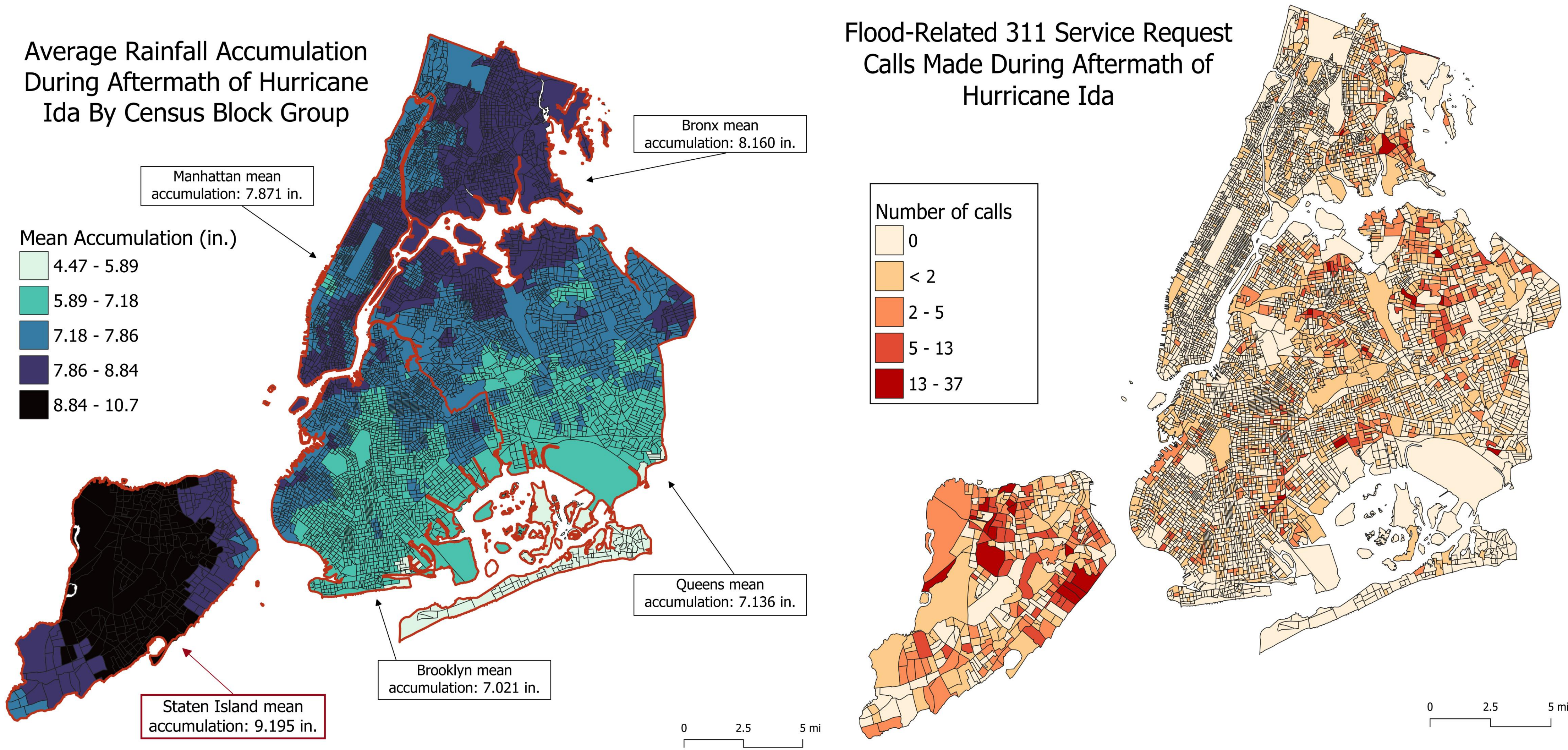
(Getty Images: Floodwaters surround vehicles following heavy rain on an expressway in Brooklyn, New York, early Thursday. Flash flooding and record-breaking rainfall were brought by the remnants of Hurricane Ida as the storm swept through the area.)

RESEARCH QUESTIONS:

How were flood-related 311 service requests distributed across New York City during the cloudbursts associated with the remnants of Hurricane Ida?

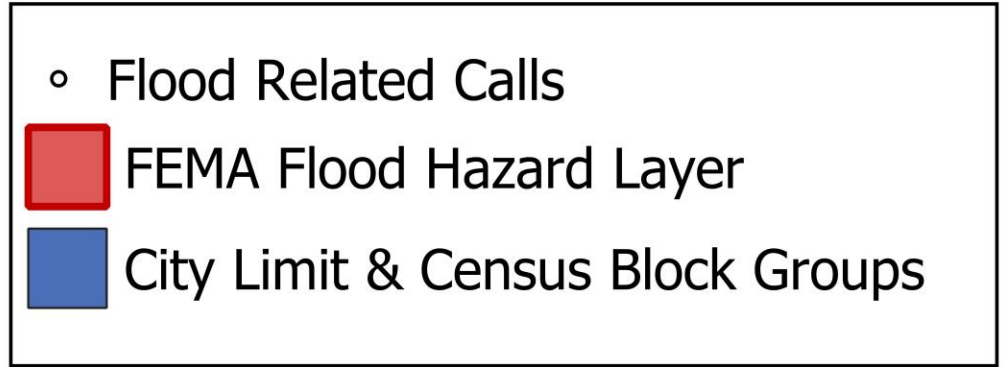
Hypothesis: Flood-related 311 service requests were made across the city due to record amounts of rainfall and socioeconomic factors, rather than coastal flooding or proximity to the coastline.

Flood Vulnerability and 311 Service Requests: A Geospatial Analysis of Ida's Impacts on New York City Communities

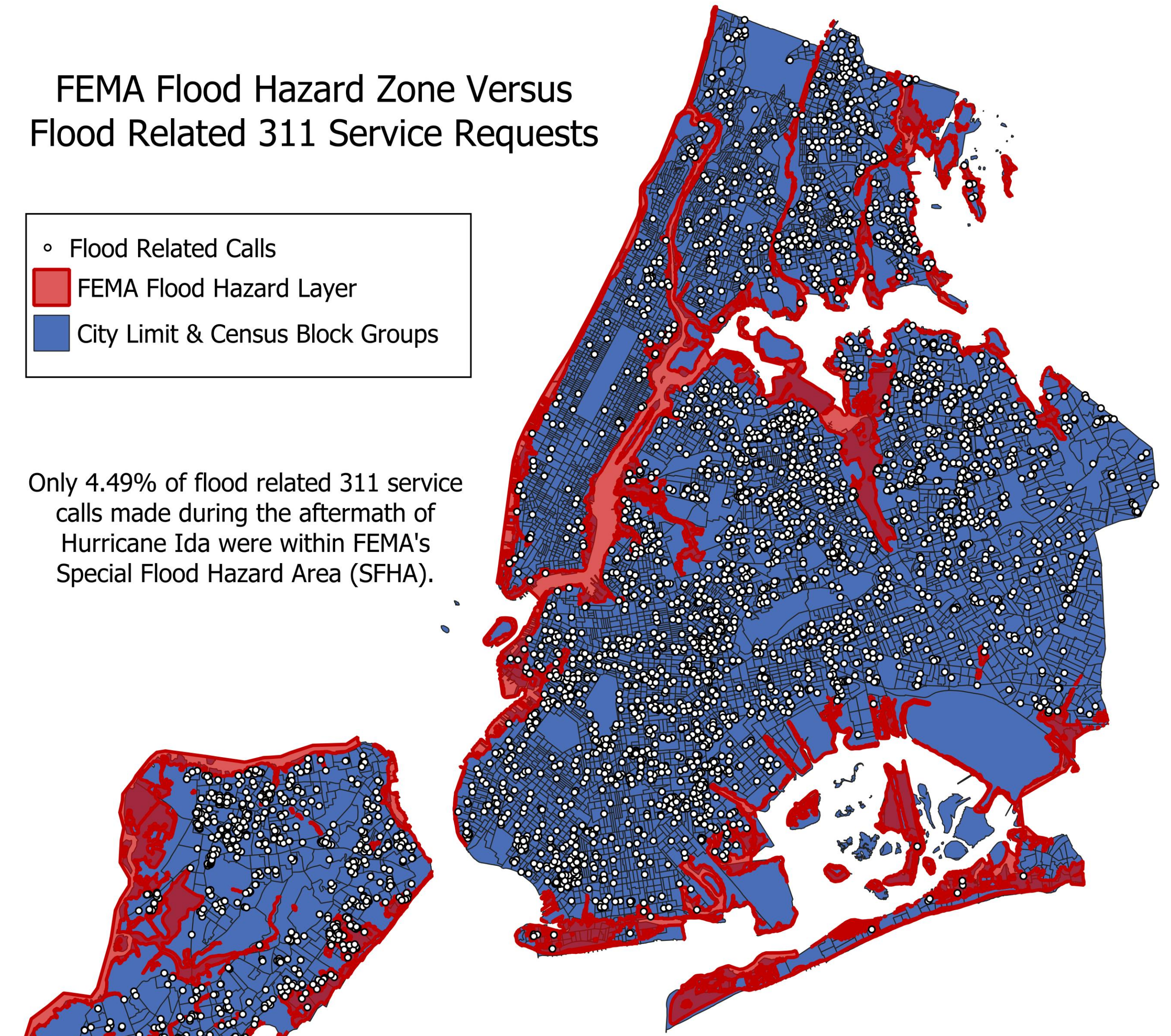


Flooding caused by Hurricane Ida impacted many communities throughout New York City.

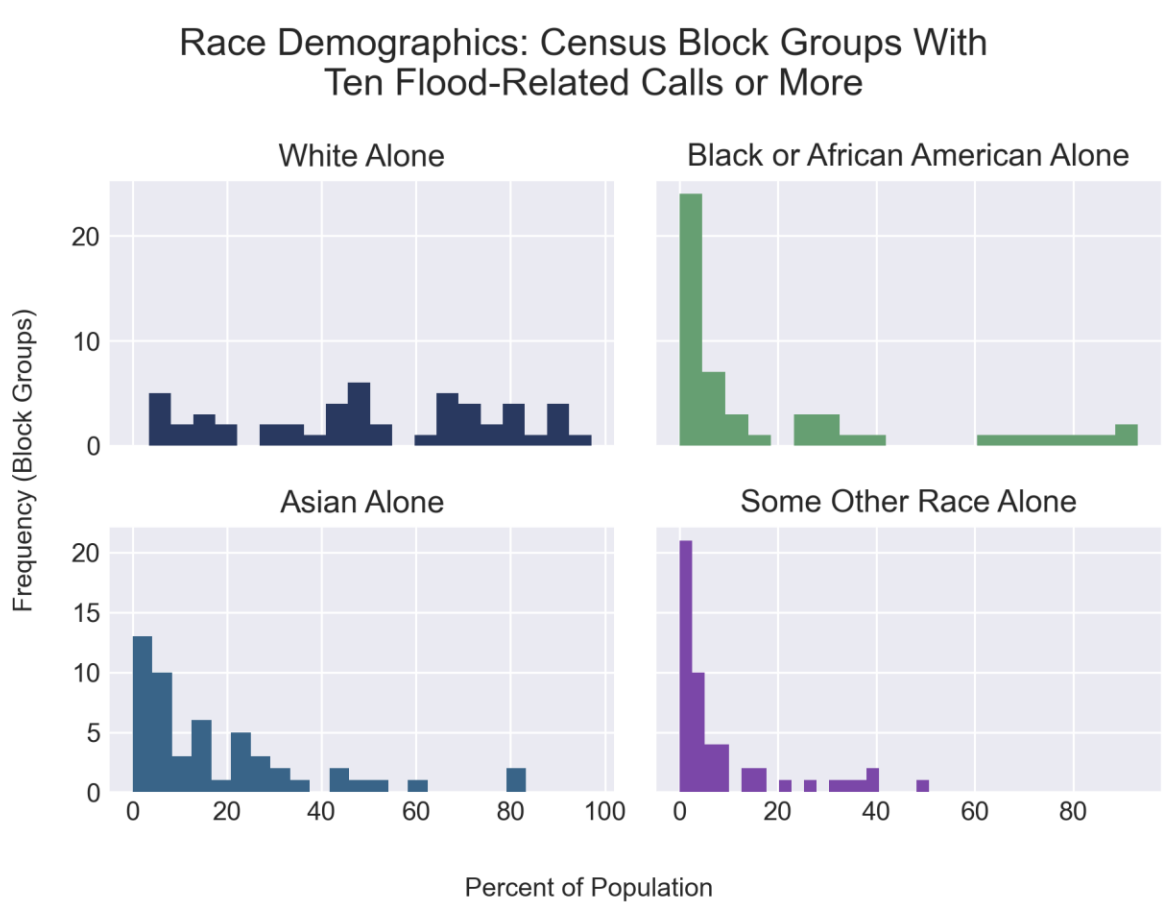
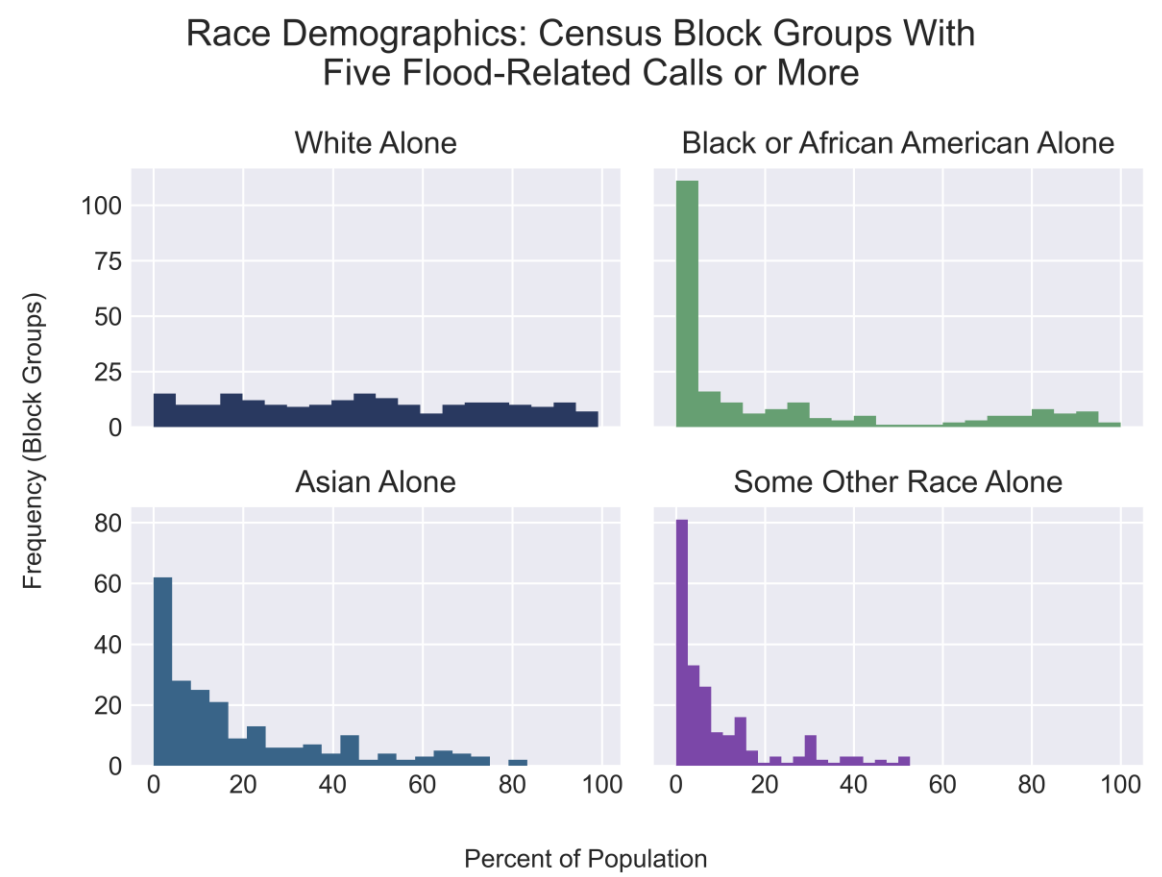
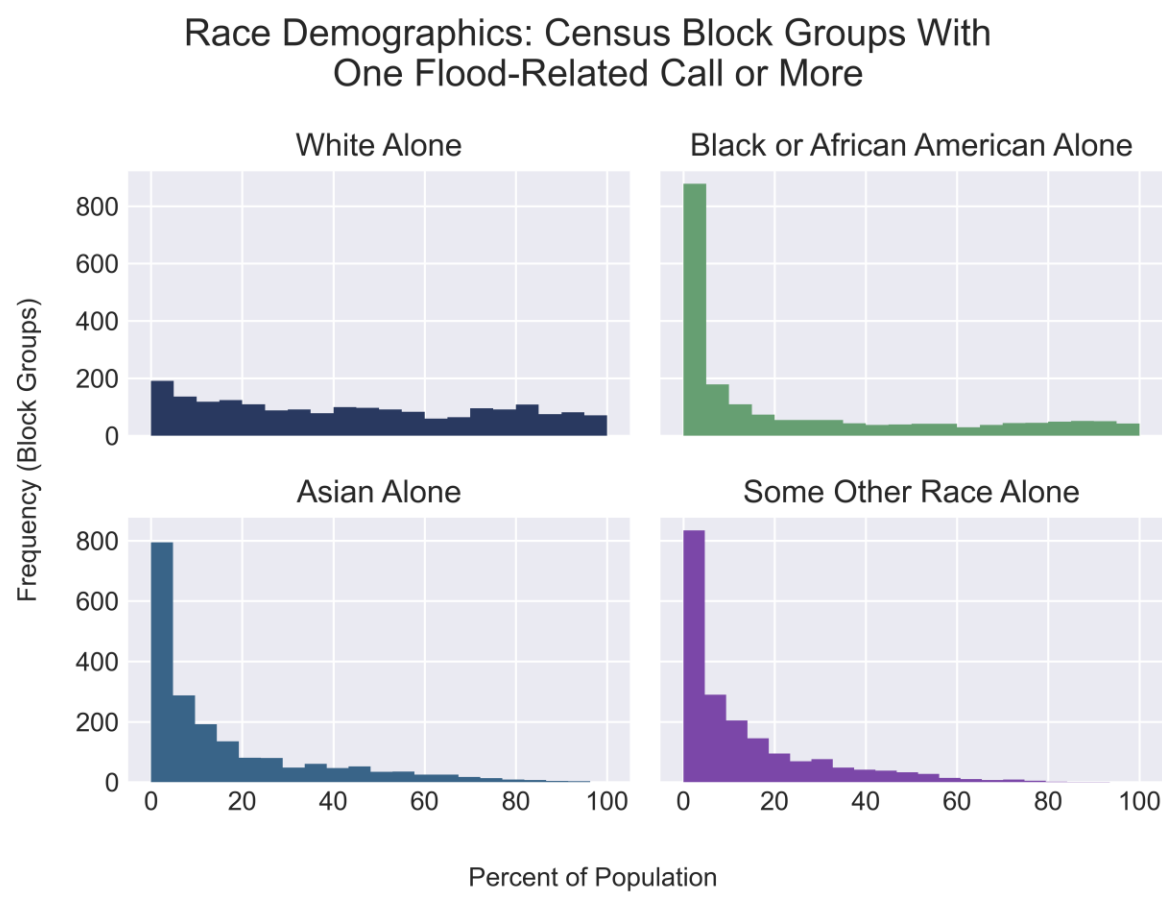
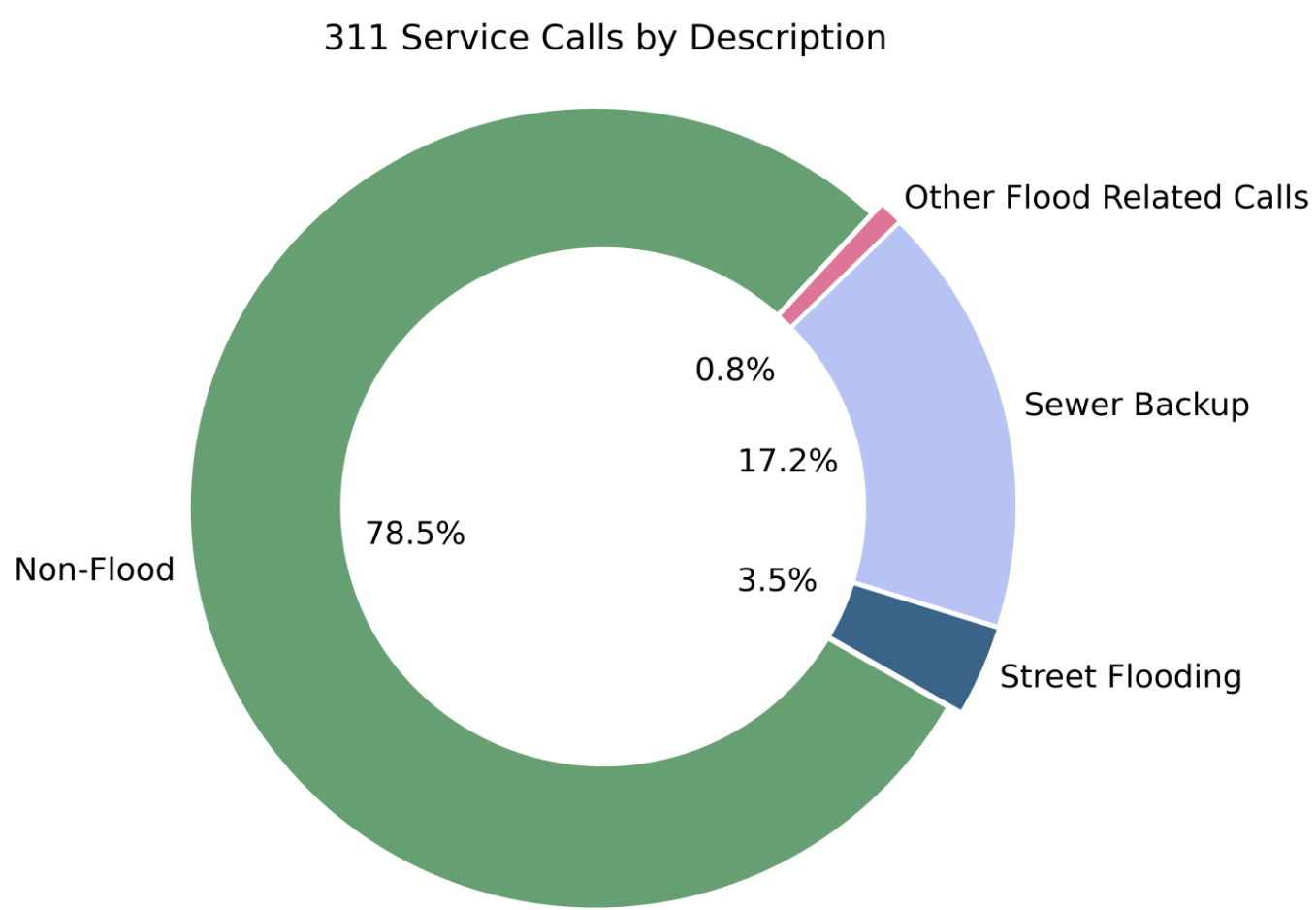
FEMA Flood Hazard Zone Versus Flood Related 311 Service Requests



Only 4.49% of flood related 311 service calls made during the aftermath of Hurricane Ida were within FEMA's Special Flood Hazard Area (SFHA).



NOTE: While 311 call data is extremely useful, communities who suffered life-threatening flooding during the aftermath of Ida would not be represented by 311 data because they would have called 911, if they had the chance to call at all.



ORG3A



METHODS:

Download Data & Format

Rainfall Radar Data Analysis

311 Service Request Analysis

Census Demographics

FEMA Flood Hazard Layer

Detailed description of methods used can be found by scanning the QR code below.

All analyses were made using QGIS version 3.22 and Python 3.10

CONCLUSIONS:

- While in general, flood vulnerability seemed to follow rainfall distribution, the areas where that pattern breaks are the most important to study.
- The term "flood vulnerability" is defined by, in this case, the number of flood-related service requests made.
- The vast majority of 311 service calls made relating to flooding were outside of FEMA's Flood Hazard Zone, meaning residents aren't required to buy flood insurance and often are unaware of any flood risk.
- Despite being thought of as only a coastal issue, flooding and flood vulnerability is a very multi-faceted issue that affects many areas of NYC.

Supplementary information, visualizations, and sources.

